ZZ–H–0141a

2 AUGUST 1951

EMERGENCY FEDERAL SPECIFICATION

HEELS; RUBBER

In the interest of conservation of certain strategic and critical materials, namely, rubber, this specification was approved on the above date by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies, and shall become effective immediately as an Emergency Specification in the purchase of the commodity covered by it, where it may be considered a suitable alternate for materials covered by Federal Specification ZZ–H–141a.

1. SCOPE AND CLASSIFICATION.

1.1. *Scope.*—This specification covers heels made of natural rubber, synthetic rubber, or a mixture of these materials.

1.2. *Classification.*

1.2.1. *Grade and types.*—Rubber heels covered by this specification shall be of one grade and of the following types, as specified in the invitation for bids:

Type I.—Half heels
Type II.—Whole heels
Type III.—Junior and Cuban heels

1.2.2. *Color.*—Rubber heels covered by this specification shall be of the following colors, as specified in the invitation for bids:

Color No. 1.—Marking black
Color No. 2.—Non-marking black
Color No. 3.—Tan
Color No. 4.—Neutral
Color No. 5.—White

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS.

2.1. The following Federal Specification, of the issue in effect on date of invitation for bids, shall form a part of this specification;

ZZ–R–601—Rubber Goods; General Specifications (Methods of Physical Tests and Chemical Analyses).

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from GSA Regional Offices.)

2.2. Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS.

3.1. *Bid samples.*—Unless otherwise specified in the invitation for bids, one pair of each type and color of heels shall be furnished with the bid to determine compliance with this specification as to design, shape, size, nonmarking quality, color, finish, and pliability. In all

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respects not specifically covered by this specification, the heels delivered shall be similar and equal to the bid samples.

3.2 *Material.*—Heels shall be made from a compound of natural rubber, synthetic rubber, or mixture of the two. Materials used in the manufacture of heels shall be of good quality free from substances that might injuriously affect the quality of the heels.

3.3. *Thickness.*—When measured as described in 4.3.1, the thickness of heels shall be as follows:

3.3.1. *Type I (half heel).*—The thickness of half heels shall be ½ inch at the back and 7/16 inch at center breast with a minus tolerance of 1/64 inch exclusive of the bevel and ridges around nail holes.

3.3.2. *Type II (whole heels).*—The thickness of whole heels shall be not less than 7/8 inch at the back and approximately 11/16 inch at center breast exclusive of the bevel and ridges.

3.3.3. *Type III (junior and Cuban heels).*—The thickness of junior and Cuban heels shall be not less than 3/8 inch at back and approximately 21/64 inch at center breast.

3.4. *Nail holes.*—Each nail hole shall be provided with a metal washer securely imbedded in the heel.

3.4.1. *Type I (half heels).*—Unless otherwise specified in the invitation for bids, each half heel shall have 8 nail holes properly located.

3.4.2. *Type II (whole heels).*—Unless otherwise specified in the invitation for bids, each whole heel shall have 13 nail holes properly located.

3.4.3. *Type III (junior and Cuban heels).*—Unless otherwise specified in the invitation for bids, each junior and Cuban heel shall have not less than 7 and 5 holes, respectively, properly located.

3.5. *Abrasion resistance.*—The abrasive index shall be as indicated in 3.5.1 and 3.5.2, when tested as described in 4.3.2.

3.5.1. *Marking black heels.*—The abrasive index of marking black heels shall be not less than 55.

3.5.2. The abrasive index of nonmarking black, tan, neutral, and white heels shall be not less than 35.

3.6. *Hardness.*—The hardness of the heels 24 hours after vulcanization, at least, shall be not less than 65 nor greater than 80, when determined as described in 4.3.3.

3.7. *Air-oven aging properties.*—The abrasive index of the heels after aging shall be not less than 75 percent of that of the unaged heels (see 4.3.4 and 7.2).

3.8. *Nonmarking.*—Nonmarking heels shall meet the requirements of the test described in 4.3.5.

3.9. *Workmanship.*—The finished rubber heels shall be free from major defects that may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1. *Sampling.*

4.1.1. Unless otherwise specified in the invitation for bids, four pairs of heels of each type, color, size, and design, from each lot of 1,000 pairs or less shall be taken at random for the purpose of inspection and tests.

4.2. *Inspection.*

4.2.1. Samples of heels shall be inspected for type, color, size, design, shape, holes, defects, workmanship, packaging, packing, and marking.

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4.3. *Tests.*

4.3.1. *Thickness.*—Thickness of rubber heels shall be determined as described in Federal Specification ZZ–R–601.

4.3.2. *Abrasion resistance.*—The abrasive index of rubber heels shall be determined as described in Federal Specification ZZ–R–601.

4.3.3. *Hardness.*

4.3.3.1 *Apparatus.*—The apparatus shall be a durometer with indentor point and spring as follows:

4.3.3.1.1. *Indentor point.*—The indentor point shall be made of hardened steel. The maximum extension of the indentor pin beyond the surface of the bearing plate of the instrument shall be 0.100 (+.000,–0.010) inch. The end of the indentor pin shall be a truncated cone 0.031-inch diameter and 35±0.25° angle.

4.3.3.1.2. *Spring.*—The force of the spring on the pin shall be 56 grams when it is fully extended (dial reading of zero) and 822 grams when the end of the pin is in the plane of the bearing plate (dial reading of 100). The force of the spring on the pin and the dial reading shall be proportional to the distance the pin is pressed back from its maximum extension (0.100 inch).

4.3.3.2. *Procedure.*—The specimen shall rest on a hard, horizontal surface. The durometer shall be held in a vertical position and applied to the specimen while keeping the presser foot parallel to the surface of the specimen. The instrument shall be lowered as rapidly as possible without shock. The pressure applied shall be just sufficient to insure firm contact of the pressure plate with the specimen. The durometer indentation reading shall be taken immediately after application of the instrument so as to obtain the maximum numerical value. All readings shall be taken not less than 0.25 inch from the edge of the specimen.

4.3.4. Accelerated aging.—The specimens shall be aged in circulating air as described in Federal Specification ZZ–R–601. The temperature shall be 70°±1° C. (158°±2°F.). (See 7.2.)

4.3.5. *Nonmarking.*—The edge of the heel shall be drawn across a sheet of white paper. If only a slight mark is made which is easily rubbed off with the fingers, the heels shall be classed as nonmarking heels.

5. PREPARATION FOR DELIVERY.

5.1. *Packaging.*—Unless otherwise specified in the invitation for bids, commercial packages are acceptable under this specification.

5.2. *Packing.*—Unless otherwise specified in the invitation for bids, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance by common or other carriers for safe transportation at the lowest rate to the point of delivery.

5.3. *Marking.*

5.3.1. *Unit packages.*—Each package shall be plainly marked on the outside to indicate the type, composition, color, size, and design of heels.

5.3.2. *Shipping containers.*—Shipping containers shall be marked with the name of the commodity and the quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or purchase order.

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6. NOTES.

6.1. *Ordering data.*—Purchasers should exercise any desired options offered herein. (See 1.2.1, 1.2.2, 3.1, 3.4, 4.1.1, 5.1, 5.2, 7.2.)

Patent Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS.

7.1. The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2. *Army accelerated aging.*—Unless otherwise specified in the invitation for bids, specimens from heels purchased by the Army shall be aged in circulating air as described in Federal Specification ZZ–R–601. The temperature shall be 100°±1° C. (212°±2° F.) and the time of exposure 24±¼ hour.

7.3. Army, Navy, and Air Force.

7.3.1. *Applicable specifications and other publications.*

Federal Specifications:

NN–B–621—Boxes, Wood, Nailed and Lock-Corner.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

Military Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case Liner.

U. S. Army Specifications:

94–40645—Marking; Exterior, Domestic and Export Shipment, by Contractor.[1]
100–2—Standard Specification for Marking Shipments by Contractor.[2]

Navy Department Specification:

General Specifications for Inspection of Material.[3]

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.[3]

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[1] Applicable to Air Force purchases only.
[2] Applicable to Army purchases only.
[3] Applicable to Navy purchases only.

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Bureau of Supplies and Accounts Publication:

Navy Handbook on Packing and Materials Handling—Packaging, Packing, and Marking of Clothing, Small Stores, and Textiles.[3]

(Copies of specifications required for Army purchases may be obtained upon application to the producing agency. Both the title and identifying symbol or number should be stipulated.

(Copies of specifications required for Air Force purchases may be obtained from the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton 1, Ohio.

(Copies of MIL–STD–105 and Navy Department specifications required for Navy purchases may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y. Both the title and identifying number or symbol should be stipulated when requesting copies.

(Applicable sections of the Navy Handbook on Packing and Materials Handling—Packaging, Packing, and Marking of Clothing, Small Stores, and Textiles may be obtained upon application to the Commanding Officer, Naval Clothing Depot, Twenty-ninth Street and Third Avenue, Brooklyn 32, New York. When requesting, bidders and contractors should state the specification number and complete nomenclature of the item to be packed so that proper sections may be forwarded.)

7.3.2. *Packaging.*—Heels purchased by the Army, Navy, and Air Force shall be packaged, packed, and marked in accordance with the following detail requirements, as specified.

7.3.2.1. *Type III, junior and Cuban heels.*—One-hundred and forty-four pairs of heels of one kind and size only shall be packaged in a fiberboard box conforming to style RSC of Specification LLL–B–631 or LLL–B–636, having a minimum bursting strength of 175 p. s. i. Arrangement of heels in the box shall be as follows:

Heels shall be placed together on end;
Twelve pairs to a row;
Four rows to a layer;
Three layers per box.

Approximate inside dimensions (inches) of the containers shall be as shown below. Actual dimensions of the same size heels may vary slightly among commercial producers; therefore the sizes of the containers must be verified by the contractor prior to packing the contents:

Heels	Sizes	Inside dimensions, inches
Cuban	1	9½ x 7 x 6.
	3	9½ x 8 x 7.
	5, 7	9½ x 9 x 7½.
Junior	J–14 thru J–17	9½ x 11½ x 9¼.

[3] Applicable to Navy purchases only.

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Any void space within the box due to the variation in size of the heels shall be filled with pads of corrugated fiberboard or equivalent cushioning material to provide a snug fit. All flaps of fiberboard containers shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps.

7.3.2.2. *Types I and II, half and whole heels.*—Heels shall be packed directly into the shipping container and will not require packaging.

7.3.3. *Packing.*

7.3.3.1. *For domestic shipment.*

7.3.3.1.1. *Type III, junior and Cuban heels.*—Two boxes of heels of one kind and size only, packaged as specified in 7.3.2.1, shall be overpacked in a fiberboard box conforming to style RSC of Specification LLL–B–631 or LLL–B–636, having a minimum bursting strength of 275 p. s. i. Each box shall be sealed as specified in 7.3.2.1.

7.3.3.1.1.1. Alternatively, two boxes of heels, of one kind and size only, packaged as specified in 7.3.2.1, shall be overpacked in a nailed wood box conforming to style 4 of Specification NN–B–621.

7.3.3.1.2. *Types I and II, half and whole heels.*—One hundred and forty-four pairs of type I heels or seventy-two pairs of type II heels one type and size only, shall be packed in a fiberboard box conforming to style FTC of Specification LLL–B–631 or LLL–B–636. Cover and body shall be assembled with not less than nine staples in each flap, spaced evenly in three vertical rows of three staples each, one row being nominally one inch from the free end of the stitched flap, the second row midway between the other two. The top and bottom staple in each row shall be approximately one inch from the edge of the flap. Each box shall be strapped as specified in 7.3.3.2.4. Arrangement of the heels in the box shall be as specified in 7.3.2.1 except that type II heels shall be arranged six pairs per row. Approximate inside dimensions of the containers shall be as shown below. Actual dimensions of the same size heels may vary slightly among commercial producers; therefore the sizes of the containers must be verified by the contractor prior to packing the contents:

Heels	Sizes	Inside dimensions inches
Type I, half heels, black, tan.	7–8, 9–10	13¼ x 11¼ x 9¼.
	10–11, 11–12	13¼ x 12½ x 10¼.
	12–13, 13–14	13¼ x 13¾ x 11½.
Type II, whole heels, black.	7–8, 9–10	12 x 11¾ x 9¾.
	10–11, 11–12	12 x 12¾ x 10¼.
	12–13, 13–14	12 x 13¼ x 10½.

7.3.3.2. *For overseas shipment.*

7.3.3.2.1. *Type III, junior and Cuban heels.*—Two boxes of heels of one kind and size only, packaged as specified in 7.3.2.1 shall be overpacked in a weatherproof fiberboard box constructed and strapped in accordance with type CF or SF, grade V3S, style RSC of Specification JAN–B–108, or equivalent, or a nailed wood box as specified in 7.3.3.2.2.2.

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7.3.3.2.2. *Types I and II, half and whole heels.*—One-hundred and forty-four pairs of type I heels or seventy-two pairs of type II heels shall be arranged as specified in 7.3.2.1 within a weatherproof fiberboard or nailed wood box as hereinafter specified.

7.3.3.2.2.1. *Fiberboard box.*—Boxes shall conform to type CF or SF, grade 3, compliance symbol V3C or V3S, style FTC of Specification JAN–P–108, fabricated in accordance with all the requirements of paragraphs under E–3 of Specification JAN–P–108, except the body blank shall be slotted on the ends, with flaps butted and stitched on the outside. The cover blank shall be side-slotted with flaps stitched on the inside. Cover and body shall be assembled with not less than nine staples in each flap, spaced evenly in three vertical rows of three staples each, one row being nominally one inch from the free end of the stitched flap, the second row the same distance from the vertical corner of the box and the third row midway between the other two. The top and bottom staple in each row shall be approximately one inch from the edge of the flap. Each box shall be strapped as specified in 7.3.3.2.4.

7.3.3.2.2.2. *Nailed wood boxes.*—Boxes shall be constructed and strapped in accordance with style 4 of Specification JAN–P–106.

7.3.3.2.3. *Case liners.*—Each overseas shipping container shall be provided with a sealed case liner or prefabricated bag made of waterproof barrier material in conformance with Type L–2 or M of Specification JAN–P–125. Closures and seams shall be completely sealed with a continuous seam of minimum ¾-inch width water-resistant adhesive in conformance with Specification JAN–P–140.

7.3.3.2.4. *Strapping and closure FTC box.*—After packing the contents, the cover shall be placed on the body and secured by applying two steel straps at right angles to each other with one strap centered over the top, sides, and bottom and the other strap centered over the top ends and bottom. The longest strap shall be applied first. Flat steel strapping is preferred but round steel strapping is acceptable. The minimum gage or size of strapping shall be as follows:

	Minimum Ultimate tensile strength
Round strap:	
15 gage	100,000 pounds per square inch.
Flat strap:	
⅜ x 0.015 inch	80,000 pounds per square inch.

The seal shall provide a joint strength of not less than 75 percent of the breaking strength of the strap used.

7.3.4. *Marking.*

7.3.4.1. *Packages.*—Each intermediate fiberboard box shall be legibility stenciled or labeled with the following information:

Type
Stock Number
Nomenclature
Quantity (Number of pairs of heels)
Size
Contract No.
Name and address of manufacturer

7.3.4.2. *Shipping containers.*

7.3.4.2.1. *Army purchases.*—Each shipping container shall be marked in accordance with the requirements of U.S. Army Specifi-

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cation 100–2, except that corner service color markings shall not be applied to containers for domestic shipment.

7.3.4.2.2. *Navy purchases.*—Each shipping container shall be marked in accordance with the requirements of the latest issue of the Navy Shipment Marking Handbook.

7.3.4.2.3. *Air Force purchases.*—Each shipping container shall be marked in accordance with the requirements of U. S. Army Specification 94–40465.

NAVY INTEREST: MC S

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog

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